

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_modular_operator_random_replay.
- Best held-out TSPLIB gap: phase7_modular_operator_random_replay.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_003008_e
- started_at_local: 2026-05-20 00:30:08
- finished_at_local: 2026-05-20 01:02:42
- duration_hhmm: 00:33
- duration_seconds: 1954.167
- seed_offset: 4000
- replicate_label: e
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.213387	0.064916	0.159398	0.86775	0.56	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.888331	0.44	False	0.010203	0.770705
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.200044	0.053637	0.107576	0.0	0.4425	False	-0.012613	1.366747
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.054968	0.121317	0.844828	0.4425	False	0.00152	-0.00642

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.054968	0.121317	0.903885	0.4425	False	0.002619	0.016329
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235058	0.054968	0.121317	0.886294	0.445	False	0.003188	-0.044844

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.213387, family holdout 0.064916, combined 0.159398.
- Accepted novelty 0.86775, complexity 0.56, and adaptation efficiency -0.028701.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.888331, complexity 0.44, and adaptation efficiency 0.006515.
- Last-epoch runtime 78.9645 ms and distance evaluations 4223.5.
- Validation: surviving False, transplant delta 0.010203, positive scaffolds 1, Pareto runtime inflation 0.770705.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.84967, "gap_delta": -0.002615, "runtime_inflation": 0.770705, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.200044, family holdout 0.053637, combined 0.107576.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 54.432625 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta -0.012613, positive scaffolds 4, Pareto runtime inflation 1.366747.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 26.433505, "gap_delta": -0.013741, "runtime_inflation": 1.366747, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.844828, complexity 0.4425, and adaptation efficiency 0.014087.
- Last-epoch runtime 168.1671 ms and distance evaluations 7395.75.
- Validation: surviving False, transplant delta 0.00152, positive scaffolds 0, Pareto runtime inflation -0.00642.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 11, "complexity_score": 0.42, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.00642, "same_gap_faster": true}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.903885, complexity 0.4425, and adaptation efficiency 0.032435.
- Last-epoch runtime 68.9473 ms and distance evaluations 4116.5.
- Validation: surviving False, transplant delta 0.002619, positive scaffolds 0, Pareto runtime inflation 0.016329.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": 0.016329, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.886294, complexity 0.445, and adaptation efficiency 0.018599.
- Last-epoch runtime 101.57055 ms and distance evaluations 7393.75.
- Validation: surviving False, transplant delta 0.003188, positive scaffolds 0, Pareto runtime inflation -0.044844.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.044844, "same_gap_faster": true}

Judge Appendix

TSP Modular-Operator Discovery Suite Evaluation

Summary of Main Results

Condition	Final TSP LIB Gap	Final Family Gap	Final Transfer Gap	Surviving Candidate	Transplant Mean Gap Delta	Ablation Validation	Comments
phase7_baseline_heuristic_only	0.235058	0.054968	0.121317	No	0.0	N/A	Baseline heuristic reference.
phase7_full_solver_evolution	0.213387	0.064916	0.159398	No	0.0	N/A	No modular operator; no surviving candidate.
phase7_modular_operator_evolution	0.235058	0.046321	0.115856	No	+0.010203	Pass (small gap deltas on ablations)	Operator: restart_controller; shows positive family signal in 2 families, but fails transplant on key scaffolds; runtime inflated (0.77x slower); transplant gap increased; no survival status.
phase7_modular_operator_random_replay	**0.200044**	0.053637	0.107576	No	-0.012613	Pass (ablation shows some gap delta but mixed)	Operator: scaffold_selector; best TSPLIB gap; positive transplant on 4 scaffolds; runtime inflation high (1.36x); fails clean transplant on a scaffold; mixed family signals with some regressions. No surviving candidate.
phase7_modular_operator_diversity_residual_replay	0.235058	0.054968	0.121317	No	+0.00152	No (ablation shows no effect)	Operator: acceptance; no family improvements; no transplant success; disabling no effect; no survival.
phase7_modular_operator_compression_pressure	0.235058	0.054968	0.121317	No	+0.002619	No (no ablation effect)	Operator: perturbation; no family improvements; no clear transplant gains; no survival.
phase7_modular_operator_pareto_selection	0.235058	0.054968	0.121317	No	+0.003188	No (no ablation effect)	Operator: perturbation; same as above in structure; no ablation or family signal; no survival.

Conservative Interpretation and Prioritization

Main Transfer Evidence

- The best transfer gap is ****lowest**** (best) for phase7_modular_operator_random_replay condition (0.107576).
- The best held-out TSPLIB gap also belongs to the same condition (0.200044).
- Family holdout gaps marginally better for modular operator evolution (0.046321) but with worse TSPLIB gap (0.235058).

Transplant Assessment

- The phase7_modular_operator_random_replay operator (scaffold_selector) shows **positive** transplant mean gap delta (-0.012613) over multiple scaffolds and 4 positive scaffolds (out of 5 tested).
- The restart_controller operator in phase7_modular_operator_evolution condition shows **positive** transplant gap delta (+0.010203), indicating worse performance after transplant.
- Perturbation and acceptance operators have near-zero or slightly positive transplant gap deltas (worse), and low or zero positive scaffold counts.

Ablation Checks

- Only two operators show ablation survival with small positive or zero gap deltas:
- **restart_controller** operator: small ablation effect (gap delta 0.002615).
- **scaffold_selector** operator: ablation shows gap increases when disabled or flattened selector variant (gap delta 0.0137); shuffled selector variant improved gap by -0.0087.
- Perturbation and acceptance operators fail ablation validation as disabling causes no performance drop.

Pareto and Runtime Tradeoffs

- The scaffold_selector operator has largest runtime inflation (1.366747), which is significant for practical use.
- restart_controller has runtime inflation ~0.77x (slower).
- Perturbation and acceptance operators show minor runtime changes (near zero or slight reduction).
- No operator improves gap with faster runtime (no "better gap same runtime").

Survival

- No operator survived overall modular scaffold and family holdout tests to be accepted as a surviving candidate.
- Though scaffold_selector and restart_controller have some positive signals, they fail transplant robustness or runtime inflation criteria.

Claims Supported by Data (Conservative)

1. **No modular-operator discovery resulted in a surviving candidate by scaffold and family holdout criteria.**
2. The **scaffold_selector** operator (phase7_modular_operator_random_replay) shows:
 - Best held-out TSPLIB gap (0.2000) and best transfer gap (0.1076) among conditions.
 - Ablation validations indicate some effect.
 - Positive transplant mean gap delta (-0.0126, improvement) and 4/5 scaffolds positive.
 - However, high runtime inflation (1.37x) and failure on clean transplant on one scaffold, preventing survival.
 - Demonstrates a reusable structure-to-scaffold mapping concept that improves performance on held-out instances conservatively.
3. The **restart_controller** operator (phase7_modular_operator_evolution) exhibits:
 - Some family holdout improvement (-0.0074 mean gap delta).

- Ablation confirms a small but consistent effect.
 - Fails clean transplant on 2 excluded scaffolds and shows runtime overhead (~0.77x).
 - Transplant gap worsened (+0.0102), indicating less robust transfer.
 - Not a surviving candidate.
 - Operator effectively regulates restart pressure for exploration but lacks transfer robustness.
4. The **perturbation and acceptance operators**:
- Show no meaningful transfer or family holdout gap improvements.
 - Ablation disables had no detrimental effects, thus no robust operator effect.
 - No surviving candidate.
5. The **full solver evolution** improved training but not transfer or held-out test, no modular structure, no operator survival.

Conclusion

- Despite some promising partial signals in the **scaffold_selector** and **restart_controller** operators, **none** pass modular operator survival checks across scaffold transplant and family holdouts.
- The main transfer gap improvements appear modest (around 1-3% absolute reduction in family gaps) but come at significant runtime costs and transplant fragilities.
- No operator demonstrates clear, robust reusable operator structure proven by survival and transplant.
- No claims about algorithm discovery or reusable operator component survival can be made based on results.
- The **scaffold_selector** operator is the most promising reusable modular operator candidate in this suite but requires further refinement for runtime efficiency and transplant robustness.

Summary table of interesting modular operators (no survival)

Operator Name	Type	Transfer Gap	Family Gap	Ablation Validity	Transplant Gap Delta	Runtime Inflation	Surviving?	Notes
context_scaffold_selector_bottle_neck_cluster	scaffold_selector	0.107576	0.053637	Partial	-0.012613	+1.37x	No	Best transfer gap, transplant gains, high runtime
restart_controller_stagnation_restart_perturb	restart_controller	0.115856	0.046321	Partial	+0.010203	+0.77x	No	Positive family signal, fragile transplant

Operator Name	Type	Transfer Gap	Family Gap	Ablation Validity	Transplant Gap Delta	Runtime Inflation	Surviving?	Notes
acceptance_threshold_worse_moves_with_stagnation_bonus	acceptance	0.121317	0.054968	No	+0.00152	~1.0x	No	No ablation effect
det_escape_perturbation_doublebridge_escalate	perturbation	0.121317	0.054968	No	+0.002619	~1.0x	No	No ablation / family signal

Final Recommendations

- Focus on improving **runtime efficiency** and achieving **clean transplant across multiple scaffolds** for scaffold_selector and restart_controller.
- Investigate why ablation signals are missing for acceptance and perturbation modular operators before further pursuit.
- Increase emphasis on robustness of family and held-out transfer gaps, alongside runtime tradeoffs, before declaring reusable modular operators.
- No operator is ready to be used as a modular discovery in production or claimed as an algorithm discovery.

End of report.